

Quality Requirements

The subsystem shall allow operators to configure tenant-specific branding (e.g. university logos, brand colour schemes, typography, and dedicated subdomains, etc.)

Security and Privacy Protection

Access to the tenant-specific branding should only be accessed by the tenant themselves by using their username and password, to protect the privacy of the university. Their branding should also only be able to be modified by someone with the username and password.

To test this, create a new account and try to access with the incorrect username or password. This would be done with multiple times with different usernames and passwords to verify that only the correct combination of username and password work to access the account.

Performance

When a user requests access to their tenant-specific branding, the system should respond within 2 seconds of the user's request.

This can be tested by using the new account created to test the security, by timing how long it takes for a logo, or colour scheme takes to load after the request. Do this multiple times to calculate the average time taken for a request to be fulfilled. This will verify that the system responds within a maximum of 2 seconds.

Reliability

The platform operator subsystem should operate continuously, only failing at most 3 times a year with a maximum time of 100 hours of the year, allowing for universities to have almost constant access to their individual brandings. However, if the subsystem does fail, it should recover within at least 10 hours of the initial failure.

To test this, a test environment should be built to simulate when multiple universities would be using the system at the same time. In this environment, multiple different types of tests can be carried out to find the breaking points of the system. This includes a stress test where the system is pushed past what would be the normal operating limits of the subsystem. And endurance test can also be done to find how long the system can run before the first failure. This measurement can be used to calculate how many failures would occur in a calendar year. Once the system fails in these tests, the time taken to get the system up and running again can be measured. This all verifies that the system would not fail more than 3 times a year with a maximum of 100 hours in that year and that the recovery time is at least 10 hours.
(GeeksforGeeks, 2019)

Scalability

The subsystem should be able to accommodate the specific branding for all 170 different universities in the United Kingdom. It should be able to process more accesses during the August/September time, when university faculty are preparing for the new academic year.

The test environment created to test the reliability of the subsystem can also be used to test the scalability of the subsystem. In the test environment, measure how long it takes to respond to

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the multiple requests and calculate the difference to how long it took for only one request.
(GeeksforGeeks, 2019)

GeeksforGeeks (2019). *Reliability Testing Software Testing*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/software-testing/software-testing-reliability-testing/> [Accessed 8 Oct. 2026].